

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

21st November 2024

SP5 - Investment and Finance

**Time allowed: 3 Hours 15 Minutes (09.30 – 12.45
Hours)**

Total Marks: 100

Q1: Company A Life uses derivatives to manage its reinvestment risk.

- (i) State the reinvestment risks that A Life can manage by use of derivatives. (4)

Your CRO has been asked to prepare a note on the use case of derivatives for the purpose of hedging?

- (ii) Discuss the risks that the company is exposed to will you put in your report to the CRO on the use of derivatives for the purpose of hedging? (9)

- (iii) What are the drivers for CIO to use FRA (Forward Rate Agreement) to hedge reinvestment risk? (4)

[17 Marks]

Q2: Risk team of Care4U life company publishes the monthly ULIP (Unit Linked) fund performance fact sheet for the customers.

List the key areas that would be covered in a fund factsheet which will help customer understand their chosen fund.

[4 Marks]

Q3:

- (i) What is a "long-only, non-leveraged strategy"? (2)

You have just joined family office of ForProfit Bank. One of the Independent Director has suggested, office should explore setting up an absolute return fund.

- (ii) Why would Independent Director have suggested such an investment opportunity? (8)

- (iii) Which kind of absolute return fund would be most suitable for this family office? Discuss the factors that you would consider while suggesting the type of absolute return fund? (7)

[17 Marks]

Q4:

- (i) In Country Paradise, company ABC is a Life and Health Insurance company. The company writes Protection, Non Par savings, Participating, ULIP and health plans for the past 10 years. During a reinsurance strategy meeting, New Business head has suggested that if company retains more, then its profitability will improve. You are the Head of ALM team and CRO has asked for an impact of higher retention on the ALM position of the company. Discuss the points that you will consider while making your report. (12)

- (ii) Your report was tabled in the next reinsurance strategy meeting. NB Head suggested that he has read about "Portfolio Theory" which will help mitigate some of the risks that have been pointed out.

- I. Discuss key aspects of "Portfolio Theory". (5)

- II. List the reasons why the theory may not be useful. (4)

[21 Marks]

Q5:

- (i) State the formula that you would use to calculate a capital value index series allowing for chain linking. Define all of the terms that you use. (2)

- (ii) Suppose that an index is to be calculated to measure the level of capital values in an emerging market in which there are currently only two shares, X & Y. At the base date of Index, there are 200X shares each priced at 4.21 and 500 Y Shares each priced at 6.11.

(a) What value is required for the index divisor if the base value of the index is to be 1,000? (2)

(b) At the close of play on the final trading day at the end of the first month, the prices of X and Y have changed to 6.68 and 5.52 respectively. What is the value of the index at that date? (2)

[6 Marks]

Q6: Your friend Mr. X is looking to buy a residential property. He told you that he thinks 1.65cr. Is a good price to pay for the said property. He is looking to buy this property as an investment. His CA is not too happy with this decision as it will impact his net of tax returns.

(i) State the factors that needs to be considered while maximizing net of tax returns. (4)

(ii) One of your other friend commented that Mr. X is getting swayed by "Anchoring and adjustment" while coming up with a price. Discuss what your friend means by the above statement and also discuss what other aspects could be applicable in friend's case? (8)

[12 Marks]

Q7: To boost business, CIO has proposed to launch a ESG fund. He suggest that this fund can cater to ESG conscious investors/customers and will help company grow its ULIP topline. You have been asked to work on a detailed proposal and submit in Investment Council for discussion. Discuss what all aspects would you consider in your proposal.

(i) What is ESG ? (3)

(ii) What are the main reasons for it to gain traction in investment domain? (4)

(iii) You have been asked to work on a detailed proposal and submit in Investment Council for discussion. The proposal should cover areas related to ESG criteria to be considered, investment strategy, risk management, regulatory compliances to taken, areas related to fund performance, sales strategy among others.

Discuss your proposal. (16)

[23 Marks]
